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Automatic text readability assessment for educational content based on graph representation learning

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Readability assessment of educational content helps ensure that texts are understandable for learners with different reading abilities. However, traditional models mainly rely on basic language features and often miss deeper patterns in organizing and expressing information. While rapid advancements in different Artificial Intelligence subfields, especially modern text embedding methods, have significantly improved readability assessment tools, existing models still need improvement. This is particularly true for educational texts, where continuous scoring is more effective than fixed-level classification to address diverse learner needs. This study proposes a graph-based method for readability assessment using Graph Convolutional Networks (GCNs) and a novel graph construction technique to represent textual structures. It models textual complexity by incorporating syntactic dependencies and assigning edge weights based on the part-of-speech tags of intermediate words. Additionally, Bayesian Optimization is used to fine-tune hyperparameters and graph construction configurations, improving the robustness and accuracy of the final model. Tested on the CLEAR dataset, the proposed method achieved an R^2 score of 0.9729. Also, the proposed method was evaluated on a classification-based dataset, and the predicted scores in the same classes were meaningfully close. The results confirm the effectiveness of the proposed method in modeling text complexity and producing accurate readability scores for educational purposes.

Keywords Readability assessment, Educational content, Natural language processing, Graph convolutional networks, Bayesian optimization

Automatic readability assessment is the quantitative evaluation of how difficult a text is to read and comprehend for a specific target audience[1]. This evaluation is especially critical in education, as texts must be accessible to learners with varying reading abilities. Even though students are paying more attention to other types of media, reading is still a key part of education because textbooks and written materials are still widely used in schools. Therefore, readability assessment is important in education to ensure that texts are appropriately challenging and accessible for all learners[2].

Conventional readability metrics, such as Flesch-Kincaid or Gunning Fog, depend on surface-level features like word length and sentence count, neglecting to account for the syntactic complexity or specialized vocabulary common in educational texts[3,4]. Historically, these static formulas were effective but typically fail to model

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hidden complexity factors of text. For example, a text with short sentences but dense scientific jargon might be assessed as 'easy,' misleading educators about its true difficulty level.

Advancements in machine learning (ML) and natural language processing (NLP) have led to new models for assessing the readability of text[5,6]. Text embedding techniques, deep learning architectures, and ensemble models are crucial. Despite these advancements, evaluating the readability of educational texts continues to be complex due to their diverse vocabulary and syntactic structures and the necessity to accommodate varied learner demographics across languages. Moreover, although many approaches assume readability as a classification task, a continuous scoring framework better reflects individual learners' comprehension abilities.

Despite recent advances in neural readability assessment, most existing approaches rely primarily on sequential representations or surface-level features and do not explicitly model syntactic structure and part-of-speech interactions within text. As a result, important linguistic relationships relevant to readability are often underutilized. This study addresses this gap by introducing a graph-based readability assessment framework that explicitly integrates POS tagging and dependency-based syntactic structure.

This study proposes a graph-based method for automated readability assessment, employing Graph Convolutional Networks (GCNs) and part-of-speech (POS)-based graph construction to overcome existing limitations. In contrast to traditional feature-based approaches, the proposed method transforms educational texts into graph representations in which words are modeled as nodes and syntactic dependencies as weighted edges. This method is distinctive as it examines the POS tags of the words between two nodes to calculate edge weights and assess sentence complexity. Furthermore, Bayesian Optimization is utilized to optimize coefficients that affect the influence of POS tags on graph weights and the model's hyperparameters.

This study makes important contributions to automatic readability assessment by introducing a new method for creating graphs that use information from POS tagging and sentence structure (dependency) analysis. This graph-based representation enables the utilization of GCNs to model the interrelations of words within sentences. Experiments on the CLEAR dataset and an additional dataset show that the proposed method consistently outperforms traditional feature-based approaches for predicting text readability. The main contributions of this study are summarized as follows:

- We propose a graph-based framework for automatic readability assessment of educational texts using Graph Convolutional Networks (GCNs).
- We introduce a POS- and dependency-informed graph construction method to capture syntactic complexity more effectively.
- We integrate contextual word representations from BERT with graph-based modeling for readability prediction.
- We apply Bayesian Optimization to jointly tune model hyperparameters and graph construction coefficients.
- We validate the proposed approach on the CLEAR dataset and an additional dataset, demonstrating improved performance over strong baselines.

This article is structured as follows: Section "Related works" reviews related work on readability assessment and graph-based NLP. Section "Proposed method" explains the proposed framework, covering text representation, model design, optimization, and experimental setup. Section "Results" presents the experimental results, Section "Discussion" discusses the findings, and Section "Conclusion" concludes the paper.

Related works

Numerous research studies have analyzed AI-driven text readability and difficulty classification in various languages. Initial techniques for evaluating text readability and complexity predominantly depended on statistical and feature-oriented methods. For example, Schwarm and Ostendorf[7] used a combination of Support Vector Machines (SVMs), statistical language models, and traditional readability formulas to classify text difficulty. They obtained recall reaching up to 75% and 79%, significantly outperforming traditional measures like Flesch-Kincaid and Lexile. Heilman et al.[8] enhanced readability prediction by incorporating grammatical and lexical features from syntactic parses and achieved a correlation of 0.767, RMSE of 2.23, and adjacent accuracy of 44% on held-out test data for predicting grade-level reading difficult. Similarly, Aluisio et al.[9] developed an SVM-based approach to assess text complexity in Brazilian Portuguese, aimed at simplifying content for low-literacy readers. Their system (SIMPLIFICA) uses classification, regression, and ranking to help writers create text suitable for various reading levels. The method attained an F-measure of 0.913 for advanced texts, 0.732 for basic texts, and a correlation of 0.84 utilizing all features in an SVM classification model.

De Clercq and Hoste[10] made an improvement by incorporating basic and advanced linguistic features, optimizing them using genetic algorithms, and demonstrating strong performance on texts in English and Dutch. In binary classification, their model achieved accuracies of 96.88% for English and 98.24% for Dutch. In the multiclass setup, the accuracies were 71.00% (English) and 73.62% (Dutch). Additionally, Balyan et al.[11] explored the use of NLP and ML models to predict text difficulty, focusing on linguistic features. Their method improved precision by nearly 10% compared to traditional readability metrics.

In addition, Madrazo Azpiazu and Pera[12] developed the CRAS (Cross-lingual Readability Assessment Strategy) framework. CRAS achieved up to 97% accuracy for English and an average of 90% across all languages using Random Forest, with interlanguage prediction agreement rising from 26% using traditional formulas to 85% using CRAS. Their research demonstrated that multilingual data enhances predictive accuracy, particularly for low-resource languages, by emphasizing critical factors influencing readability.

Weiss et al.[13] proposed a cross-lingual approach to readability assessment, utilizing an extensive model of linguistic complexity. The researchers developed the first multi-level German readability detector for L2 learners and evaluated it on German and English texts. Their feature-based classifiers achieved 92.06% accuracy on the

English OneStopEnglish benchmark–surpassing previous state-of-the-art–along with 74.5% (English) and 88.0% (German) accuracy on the Spotlight corpora. In zero-shot cross-lingual tests, the model reached 55.5% accuracy from English to German and 53.4% from German to English, well above the majority baselines.

Khallaf and Sharof[14] developed the first Arabic sentence difficulty classifier aligned with the Common European Framework of Reference (CEFR) levels. They analyzed diverse linguistic characteristics and neural network architectures, concluding that the fine-tuning of Arabic-BERT embeddings produced the most advantageous results. The best 3-way CEFR classification achieved an F1-score of 0.80 using fine-tuned Arabic-BERT, the binary difficulty classifier reached F1-scores of 0.94 and 0.98, and the regression model achieved up to 0.71 Spearman and 0.54 Kendall's tau-b correlation with the gold standard.

Extending the application of neural models to sentence-level readability assessment, Liu et al.[15] compared neural, hybrid, and large language models (LLMs) on both English and Chinese datasets. Their study was the first to assess LLMs for sentence-level readability and to introduce Chinese sentence-level datasets. Hybrid models, which integrate linguistic features with neural predictions, demonstrated superior performance, achieving 85.6% accuracy on the WSJ dataset and 78.2% on Chinese datasets, surpassing both individual neural models and LLMs.

Li et al.[16] presented a unified BERT-based model designed for cross-lingual readability assessment in both English and Chinese. Their BERT-FP-LBL model integrates feature projection with a length-balanced loss function to address challenges stemming from discrepancies in text length. This model obtained the highest results, with an F1 score of 92.73% on WeeBit, 99.41% on OneStopEnglish, 87.73% on Cambridge, and 78.75% on ChineseLR and Spearman correlations that matched real experts at 0.836.

Ivanov[17] investigated sentence-level complexity prediction in Russian by comparing the performance of BERT, graph neural networks, and traditional models. The fine-tuned RuBERT model achieved top performance with $R^2 = 0.80$, MAE = 0.80 for regression and F1 = 81.99% for 3-class classification, while in binary classification of sentence pairs it reached an F1-score of 98.47%, slightly outperforming the GNN which also performed strongly with $R^2 = 0.73$, MAE = 1.10, and F1 = 96.60%. Their findings indicated that optimized Russian BERT models yielded superior outcomes when integrating syntactic and lexical features, whereas graph neural networks offered a comprehensible alternative. Similarly employing graph-based methods, Varnamkhasti[18] developed a model for Persian readability classification by integrating DeepWalk-based graph embeddings with tree-based models such as Extra-Trees and Gradient Boosting. This method significantly enhanced the assessment of Persian text readability, achieving an exceptional F1 score of 0.961, thereby encapsulating the morphological and syntactic intricacies of Persian.

While graph-based methods like Varnamkhasti[18] use DeepWalk embeddings and Ivanov[17] applies graph neural networks, neither incorporates POS-based syntactic distance into edge weights, as the proposed method does. Similarly, deep learning models such as BERT[16] demonstrate strong capabilities in contextual understanding. However, they fall short in explicitly capturing syntactic structure. The proposed framework addresses this limitation through an innovative graph-based design that more effectively incorporates syntactic information.

Van Ngo and Parmentier[19] studied the relationship between sentence-level and document-level readability in French. They enhanced CamemBERT models with conventional readability attributes utilizing a dataset from children's literature.Sentence readability analysis with CamemBERT achieved F1 scores between 53% and 59% on randomly split data. Lower scores, ranging from 34.73% to 46.33%, were observed when using disjoint datasets. At the same time, Kincaid Grade Level (KGL) and Flesch Reading Ease (FRE) had strong links with how readable the texts were (for example, KGL scores went from −3.23 for level 0 to 2.76 for level 4).

Pickelmann et al.[20] presented the "Ablesbarkeitsmesser," an online platform for assessing the readability of German texts. The system integrates traditional readability metrics with BERT-based ML to deliver detailed, sentence-level assessments. Evaluated on the Deutsche Welle dataset, the BERT model achieved an accuracy of 86.95% in binary readability classification (easy vs. hard). The TextComplexityDE dataset achieved a Pearson correlation of 0.945 between predicted and actual readability scores.

Leal et al.[21] introduced NILC-Metrix, a computational tool featuring 200 metrics for analyzing the complexity of Brazilian Portuguese texts. Insights from cognitive science, computational linguistics, and psycholinguistics are used in this system. Building on the Coh-Metrix-Port project, NILC-Metrix enhances natural language processing capabilities by incorporating a broader set of metrics. Using 200 NILC-Metrix features, the method achieved a Macro F1-score of 0.704 in classifying text complexity levels–original, natural, and strong simplification–on the PorSimples corpus. Among the various feature categories, Syntactic Complexity (13 metrics) and Readability Formulas (5 metrics) yielded the best performance. Traditional readability formulas such as Flesch and Dale-Chall also demonstrated strong discriminative power. Furthermore, the study highlighted the tool's versatility through its effective application in tasks such as fake news detection, with 92% accuracy, and dementia diagnosis, achieving an F1-score of 0.82.

Ribeiro et al.[22] investigated automatic readability assessment for European Portuguese through classification and regression methods. They aimed to predict CEFR text levels using the Albertina PT-PT basis model, considering the levels' ordinal structure. Their findings show that regression methods that include class probability distributions outperform adjacent methods regarding accuracy and generalizability across multiple datasets. Their study achieved 80.07% accuracy and 97.27% adjacent accuracy in cross-validation using softmax regression with Albertina PT-PT, outperforming classification (80.02%, 96.96%) and regression (79.10%, 97.68%). On the test set, regression generalized best (49.06% accuracy, 92.81% adjacent accuracy), demonstrating the value of ordinal modeling for CEFR-level readability assessment in European Portuguese.

Despite their effectiveness, existing readability assessment methods often fail to adequately capture complex semantic and syntactic structures inherent in educational texts. Conventional feature-based approaches rely primarily on surface-level metrics, while deep learning models, although more expressive, typically depend on

large amounts of annotated data and offer limited interpretability. Recent graph-based approaches show promise in modeling structural relationships, yet their adoption in readability assessment remains sparse, particularly in the context of educational materials.

In summary, prior work on readability assessment spans traditional feature-driven models and more recent neural and syntax-aware approaches. While neural methods have improved predictive performance, they largely rely on sequential representations and do not explicitly encode syntactic dependencies between words. Consequently, the potential of graph-based representations to jointly model semantic content, syntactic structure, and part-of-speech interactions for readability assessment remains underexplored. Motivated by these gaps, this study proposes a graph-based readability assessment framework that explicitly integrates POS tagging and dependency-based syntactic information.

Proposed method

This study presents a novel graph-based method for evaluating the readability of educational texts. Unlike traditional metrics focusing on surface features such as word or sentence length, the proposed approach uses graph representation learning to involve deeper syntactic structure. This method models words as nodes and syntactic dependencies as edges, enabling a more accurate and linguistically informed text complexity analysis.

Edge weights are established based on the syntactic distance between dependent words, considering both the quantity and part-of-speech tags of intervening words. This design assumes that syntactic distance substantially affects a text's readability. The constructed graph is then used to generate embeddings based on BERT while involving graph structure.

The motivation for POS-based syntactic distance is grounded in psycholinguistic findings that sentence processing difficulty increases with the number and type of intervening constituents between syntactically related elements (e.g., subject–verb or verb–object dependencies). Longer dependency spans and structurally complex intervening material increase working memory load and disrupt incremental parsing, which are known to correlate with reduced readability[23]. In particular, intervening content words (e.g., nouns, adjectives, verbs) introduce higher semantic and syntactic integration cost than function words (e.g., determiners or conjunctions), motivating differentiated weighting by POS category.

Formally, for a dependency edge between tokens w_i and w_j, we define the POS-based syntactic distance as:

$$d(i,j) = \sum_{k=\min(i,j)+1}^{\max(i,j)-1} \alpha_{\mathrm{POS}(w_k)}, \tag{1}$$

where α_{POS} is a learnable coefficient associated with each POS category. This formulation generalizes raw linear distance by incorporating linguistically motivated penalties for intervening syntactic material. The coefficients are optimized via Bayesian Optimization, allowing the model to learn which POS categories contribute more strongly to readability difficulty in practice.

Figure 1 provides an illustrative example supporting the POS-based syntactic distance formulation by contrasting two sentences with similar lexical content but different intervening POS structures. Parentheses indicate the respective POS tags for each word in each sentence.

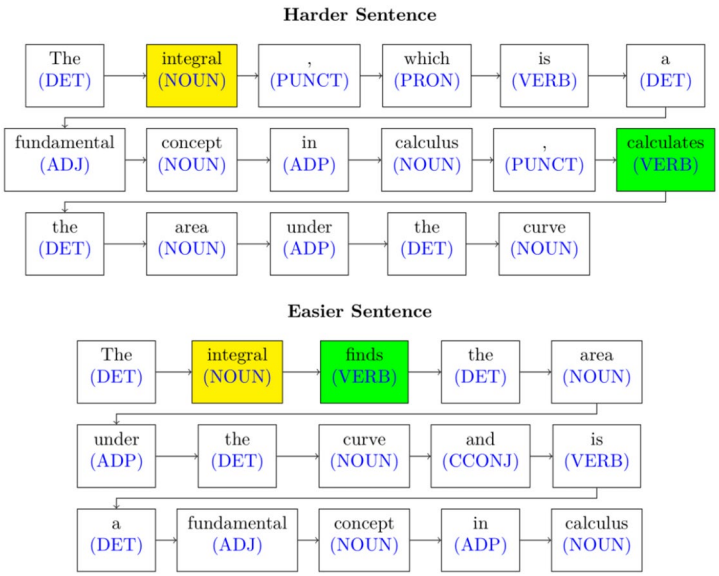


Fig. 1. Comparison of two sentences containing similar vocabulary but differing syntactic structures. The figure demonstrates the impact of POS-based syntactic distances between key terms on sentence readability.

The POS of words between two key terms (e.g., "integral" and "calculates") significantly affects text readability. In the harder sentence, the relative clause ("which is a fundamental concept in calculus") introduces additional POS elements (PRON, VERB, ADJ, NOUN, ADP) between "integral" and "calculates," increasing syntactic complexity. This nesting can hinder the flow; necessitating increased cognitive effort to comprehend the relationship between subject and verb.

Conversely, the easier sentence places the verb "finds" immediately after "integral" and connects "area" and "curve" with a straightforward prepositional phrase ("under the curve"), followed by a coordinated clause ("and is a fundamental concept in calculus"). In this sentence, POS tags between verbs and nouns preserve clear action-object relationships, reducing cognitive load and ambiguity.

As a result, POS sequences between related words influence sentence coherence and comprehension. A concise, simple POS sequence (as in the easier sentence) improves readability by minimizing syntactic breaks. In contrast, a complex, nested structure (as in the harder sentence) can hinder it, underscoring the importance of POS in text design.

Figure 2 presents the overall architecture of the proposed graph-based readability assessment framework. The architecture consists of four main stages: (1) text preprocessing using POS tagging and dependency parsing, (2) POS- and dependency-informed graph construction, (3) graph representation learning using Graph Convolutional Networks combined with contextual embeddings, and (4) readability prediction through a regression or classification layer. This modular design enables the joint modeling of semantic and syntactic information relevant to text readability. The following sections explain how to construct graphs, generate embeddings, and apply GCNs to scored readability levels. All experiments were implemented using PyTorch (v1.13) and HuggingFace Transformers (v4.31), with spaCy (v3.6) for POS tagging and dependency parsing.

Graph construction

This work creates text graphs where each word acts as a node, and the syntactic relationships between words, like subject-verb or verb-object dependencies, create edges. Edge weights are computed to capture a more nuanced representation of relationships between words. Their foundation is based on a weighted enumeration of words between two related terms, incorporating POS data. This approach ensures that the edge weights reflect the syntactic distance of the dependent terms, leading to a more accurate representation of linguistic structure. These graphs utilize intricate relationships and connections to clarify complex linguistic patterns that elevate textual complexity beyond traditional feature-based readability models.

- Nodes: Represent individual words in the text.
- Edges: Represent syntactic relationships (subject-verb, verb-object, etc.) between words.
- Edge Weights: The physical distance between two related words in a text is measured by counting the number of intervening words, weighted by coefficients based on their part of speech. For example, nouns between two words may have a greater influence on their relationship than verbs. These POS-based coefficients are treated as hyperparameters and optimized alongside other model-specific variables.

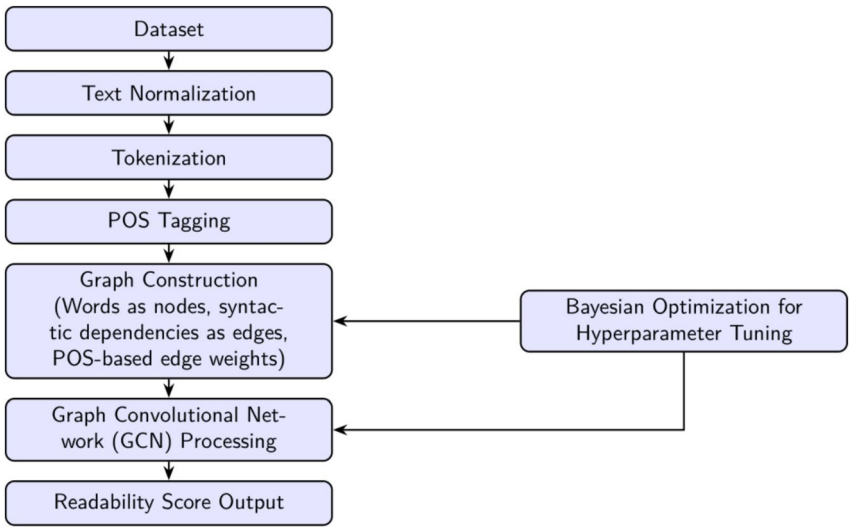


Fig. 2. Overall architecture of the proposed graph-based readability assessment framework, illustrating preprocessing, graph construction, GCN-based representation learning, and prediction stages.

Inputs: Text T, Part of speech coefficients vector C_{POS}
Output: Graph $G = (V, E)$
S: Tokenize the text into words as a vector.
S_{POS}: Part-Of-Speech tagging of the words, where S_{POSi} is part of speech of i-th word.
R: List of relationships in the text in format of triplets such as $(WordIndex_1, WordIndex_2, relationship)$.
$V = S, E = \phi$
for $(WordIndex_1, WordIndex_2, relationship) \in R$ **do** :
    $w = 0$
    for $i \in [WordIndex_1, WordIndex_2]$ **do**:
        $C = C_{POS}\ of\ S_{POSi}$
        $w = w + C$
    end for
    Add edge e with the weight w between $S_{WordIndex_1}$ and $S_{WordIndex_2}$ to E.
end for
Return G

Algorithm 1. Graph Construction Algorithm

Algorithm 1 employs syntactic structures to construct a graph representation of a text. The input text is first tokenized into words, which are graph nodes. The Arc-Standard transition system and a neural network classifier extract syntactic relationships between words, including subject-verb and verb-object dependencies[24]. POS tagging and syntactic dependency extraction were performed using the spaCy NLP library[25] (version 3.7) with the en_core_web_trf model, which implements an Arc-Standard transition-based dependency parser. In this parsing method, a trained neural model directs incremental shift-reduce operations to create dependency trees, which determine the text's syntactic structure.

After identifying syntactic relationships, they are recorded as triplets (e.g., $(WordIndex_1, WordIndex_2, relationship)$) connecting word indices. The algorithm initializes a graph with words as nodes and begins without edges. For each relationship, it creates an edge between the related words. The edge weights are calculated using the number of words between them and their part-of-speech contributions. The weights act as hyperparameters to help make adjustments to better represent the influence of the syntactic relations on readability.

Graph-based representations enable a more nuanced assessment of readability that transcends conventional lexical and statistical features by enabling deeper linguistic analysis. An illustrative example of the introduced graph structure is presented in Fig. 3.

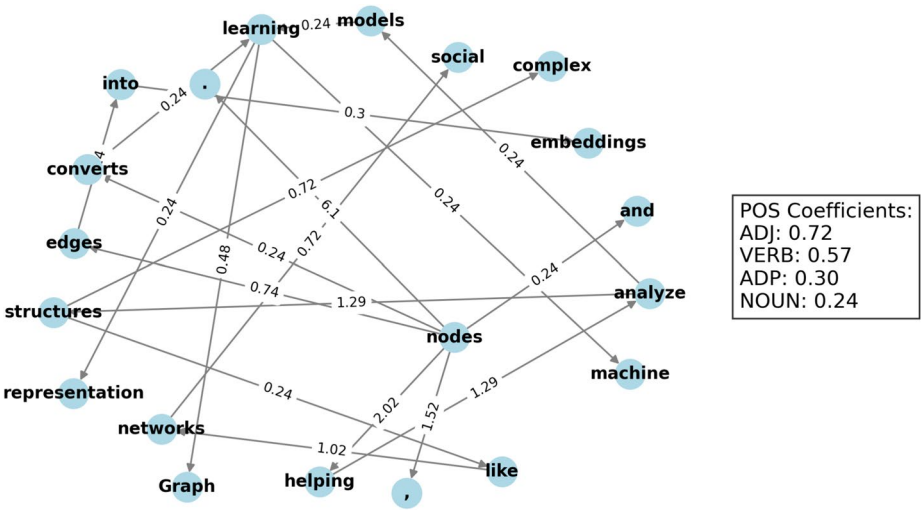


Fig. 3. Illustrative example of a syntactic dependency graph constructed from the sentence: "Graph representation learning converts nodes and edges into embeddings, helping ML models analyze complex structures like social networks.".

GCN

GCNs have been extensively applied in graph-based learning tasks due to their ability to effectively capture both global and local structural information from graph-structured data[26,27]. In this study, after constructing the graph representation of the text as described in Section "Graph construction", a GCN model is applied to perform regression aimed at assessing readability.

GCN formulation for regression

A GCN operates by aggregating features from neighboring nodes, allowing information propagation across the graph. The key update rule for a single GCN layer is given by:

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}) \tag{2}$$

In this equation, $H^{(l)} \in \mathbb{R}^{N\times d}$ indicates the node feature matrix at layer , with N nodes and d-dimensional features, $\tilde{A} = A + I$ stands for the adjacency matrix with self-loops, $\tilde{D}$ denotes the degree matrix of $\tilde{A}$, where $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Also, $W^{(l)}$ stands for the trainable weight matrix for layer and σ denotes a non-linear activation function (e.g., ReLU).

The final node embeddings from the last GCN layer are aggregated to form a graph-level representation using a readout function, such as:

$$Z = \text{MeanPooling}(H^{(L)}) \tag{3}$$

where $Z \in \mathbb{R}^{d'}$ represents the graph-level embedding used for regression.

Graph embeddings

In the proposed framework, each node in the graph represents a word in the text. The initial feature vector is derived from contextual embeddings generated by BERT (Bidirectional Encoder Representations from Transformers)[28]. Unlike static word embeddings generated by models like Word2Vec, BERT produces contextualized representations that capture the semantic and syntactic nuances of each word based on its surrounding context within the sentence[16,28]. The resulting node representations are aggregated by the GCN layers and passed to the final prediction layer to estimate text readability.

To initialize the node features, the text is first run through a pre-trained BERT model to get embeddings. Specifically, BERT tokenizes the text for a given sentence and produces a sequence of hidden states corresponding to each token. In cases where words are split into subword tokens, the embeddings of the subword units are averaged to obtain a single vector representation per word, following established practices in natural language processing. The initial node feature matrix $H^{(0)} \in \mathbb{R}^{N\times d}$ is then constructed, where each row corresponds to the BERT embedding of the respective word, N is the number of nodes (words) in the text, and d is the embedding dimension (e.g., 768 for BERT-base)[28].

While BERT provides rich node-level features, the proposed readability assessment method requires a graph-level representation of the entire text for regression. To achieve this, the Graph Convolutional Network (GCN) processes the graph across multiple layers, aggregating information from neighboring nodes according to the syntactic structure, which is encoded through edges and part-of-speech (POS)–based weights. The updated node embeddings from the final GCN layer ($H^{(L)}$) capture both the contextual semantics from BERT and the structural complexity from the graph. These are then aggregated using a readout function, which is mean pooling ($Z = \text{MeanPooling}(H^{(L)})$), to produce a graph-level embedding $Z \in \mathbb{R}^{d'}$. This graph-level embedding captures the text's overall complexity. It is then used to predict the readability score via a fully connected layer, similar to approaches that combine contextual embeddings with graph-based learning for NLP tasks.

Readability score prediction

The graph embedding Z is passed through a fully connected feedforward network to predict the readability score:

$$\hat{y} = W_{out}Z + b \tag{4}$$

where $W_{out} \in \mathbb{R}^{1\times d'}$ and $b \in \mathbb{R}$ are learnable parameters. The model is trained using a mean squared error (MSE) loss function:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{5}$$

Where, y_i stands for the expected readability score for the i-th text and $\hat{y}_i$ represents the predicted score.

The GCN model is trained using stochastic gradient descent (SGD) to reduce overfitting, with dropout and L2 regularisation. To ensure robustness, model performance is evaluated using k-fold cross-validation.

This GCN-based regression approach effectively captures syntactic and semantic complexity, outperforming traditional feature-based readability metrics by employing the structural properties of the text graph.

Bayesian Optimization (BO)

BO is a powerful and efficient method of global optimization widely used in the tuning of hyperparameters in complex ML models[29]. The proposed method of this study uses BO to optimize several critical hyperparameters, including model-specific hyperparameters such as the learning rate, number of GCN layers, and dropout rate. It also optimizes embedding-related configurations, such as the embedding dimension and initialization strategy, as well as the weighting coefficients assigned to different parts of speech during graph construction.

During its iterative hyperparameter selection process, BO constructs a probabilistic surrogate model of the objective function to improve performance on validation data[30]. The optimization process involves the following steps:

1. Get started with a limited selection of randomly sampled hyperparameters.
2. Train the model using the selected hyperparameters.
3. Evaluate performance on validation data.
4. Modify the probabilistic surrogate model.
5. Using an acquisition function, choose the next set of hyperparameters (for example, Expected Improvement or Upper Confidence Bound).
6. Iterate until convergence or a predetermined maximum number of iterations is achieved.

BO uses a Gaussian Process (GP) to model the function mapping hyperparameters to validate performance. Based on previous observations, GP computes a posterior distribution of the objective function, facilitating the effective selection of advantageous hyperparameters.

The acquisition function determines the next hyperparameter set:

$$\theta_{next} = \arg\max_{\theta} a(\theta) \tag{6}$$

where θ represents the hyperparameters being optimized.

Bayesian Optimization is employed to extensively investigate the hyperparameter space of both the GCN model and the graph construction process, aiming to determine the optimal configuration and thereby improve the performance of the readability assessment.

Bayesian Optimization was implemented utilizing a Gaussian Process (GP) as the surrogate model and the Expected Improvement (EI) as the acquisition function. The initial quantity of random points was established at 10, with the total number of iterations fixed at 35. The optimization goal was to maximize the mean R^2 score obtained from 10-fold cross-validation, ensuring consistent and reliable performance of the chosen hyperparameters across various data subsets.

All hyperparameters and implementation details used in our experiments are summarized in Table 1. Unless otherwise stated, the same configuration is used across all datasets and cross-validation folds.

Evaluation

To evaluate proposed method, the CLEAR (CommonLit Ease of Readability) Corpus[31], comprising approximately 5,000 excerpts of reading passages, was employed. Figure 4 illustrates the distribution of Motion Picture Association of America (MPAA)-based content ratings assigned to the texts within this dataset. The majority of the passages (n=3,706) are categorized as General (G), signifying their suitability for a broad spectrum of readers. A smaller portion (928) falls into the Parental Guidance (PG) category, implying that some content may require discretion. Only 87 texts are classified as PG-13, with only three rated Restricted (R). These ratings were determined through a screening process that included automated filtering and manual review to ensure the content was appropriate for educational purposes.

To further evaluate the generalizability of the proposed method–particularly across different languages–the dataset introduced by Varnamkhasti[18] was incorporated.The dataset consists of Persian educational texts and

Component	Setting
Pretrained encoder	BERT-base (uncased)
Embedding dimension	768
GCN layers	2
Hidden dimension (GCN)	256
Activation function	ReLU
Dropout rate	0.5
Learning rate	2×10^{-5}
Optimizer	AdamW
Batch size	16
Training epochs	20
Weight decay	0.01
Early stopping patience	5

Table 1. Implementation details and hyperparameter settings used in all experiments.

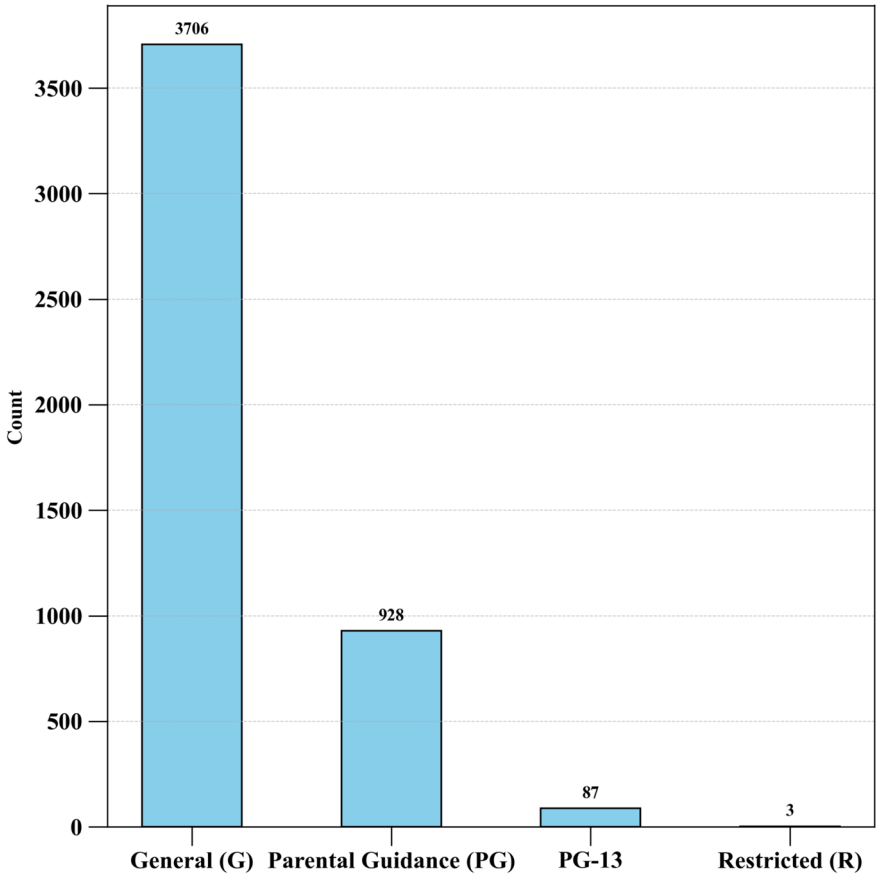


Fig. 4. MPAA-Based Content Ratings Distribution.

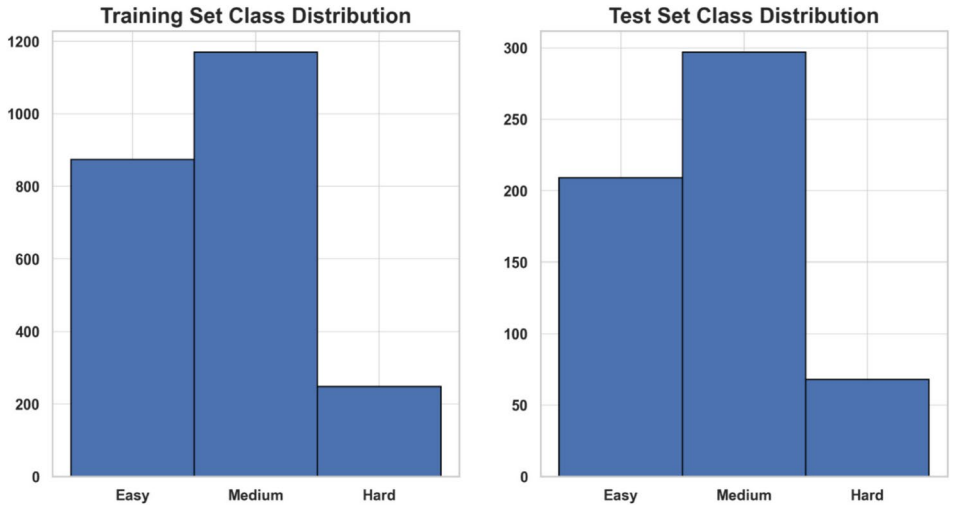


Fig. 5. Readability Classes Distribution in Varnamkhasti Dataset[18].

was built for assessing readability as a classification task. This research applies the dataset to explore whether the model could consistently give meaningful readability scores to texts from the same class and whether comparable texts were given comparable scores. For this aim the model re-trained on the dataset itself. Figure 5 illustrates the distribution of readability classes in the Varnamkhasti dataset, underscoring the range of complexity levels present in the Persian texts used for cross-lingual evaluation.

We employed a standard 10-fold cross-validation protocol, in which the dataset was randomly partitioned into ten equally sized folds. For each fold, nine folds were used for training and one fold for testing. Bayesian

Dataset	RMSE	MAE	MAPE	R^2-score
Train	0.1311	0.1012	15.98%	0.9839
Test	0.1668	0.1406	17.09%	0.9729

Table 2. Regression Results of the proposed method.

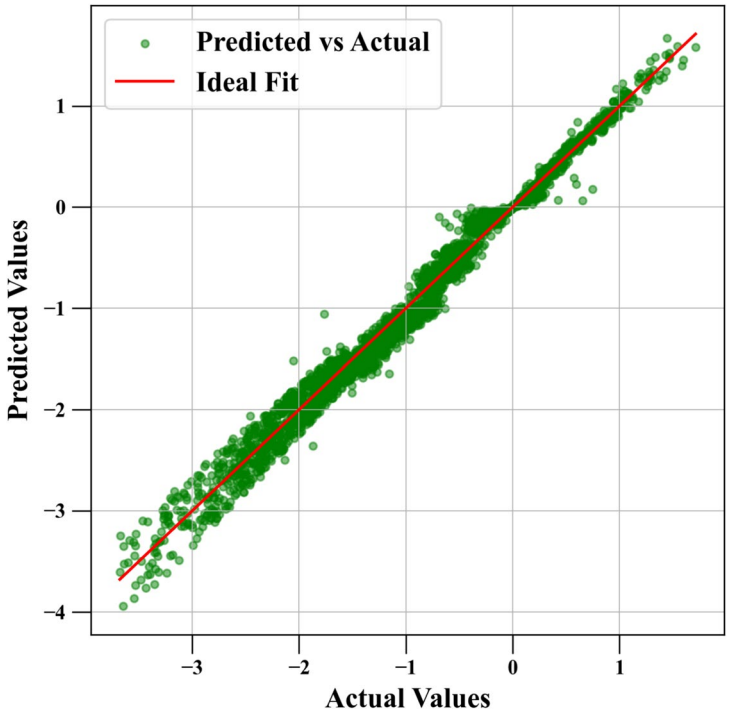


Fig. 6. Scatter plot comparing predicted readability scores from the proposed GCN-based model with actual target values for the training set of the CLEAR dataset.

Optimization was performed exclusively on the training portion within each fold, using an internal validation split, and the optimized hyperparameters were then applied to the corresponding test fold. This nested evaluation procedure ensures that no information from the test folds was used during hyperparameter tuning.

Results

This section presents the performance evaluation of the proposed graph-based model for readability assessment. The model is assessed as a regression task using the CLEAR dataset, which provides sentence-level readability scores. Model accuracy is evaluated using key performance metrics, including the R^2 score, mean absolute error (MAE), and root mean squared error (RMSE). The effect of graph construction parameters and hyperparameter optimization on model performance is examined, revealing the method's robustness and adaptability across readability levels.

Table 2 presents the regression results for the proposed graph-based readability assessment model on both the training and test datasets. On the training set, the proposed method obtained an RMSE of 0.1311 and an MAE of 0.1012, along with a high R^2 score of 0.9839. R^2 remains high on the test set at 0.9729, while RMSE and MAE decrease slightly to 0.1668 and 0.1406, respectively. Models performed slightly worse when moving from training to unseen data, but their ability to generalize proved their strength. The Mean Absolute Percentage Error (MAPE) values of 15.98% (train) and 17.09% (test) further confirm the model's robustness in predicting readability scores with minimal relative error.

A comparison of the predicted readability scores of the training set and their actual values is shown in Fig. 6. There is excellent agreement between expected and actual values based on the closely aligned data points along the diagonal. Table 2 illustrates the model's training accuracy visually, confirming the quantitative metrics.

A Q-Q plot of the training data is shown in Fig. 7. A strong alignment between the points and the reference line indicates a relatively normal distribution for the residuals. The normality of residuals is essential to determining the reliability of a regression model's statistical inferences.

Overall, the analyses of Table 2, Figs. 6, and 7 collectively demonstrate that the proposed graph-based readability assessment model effectively predicts readability scores with high accuracy and generalization

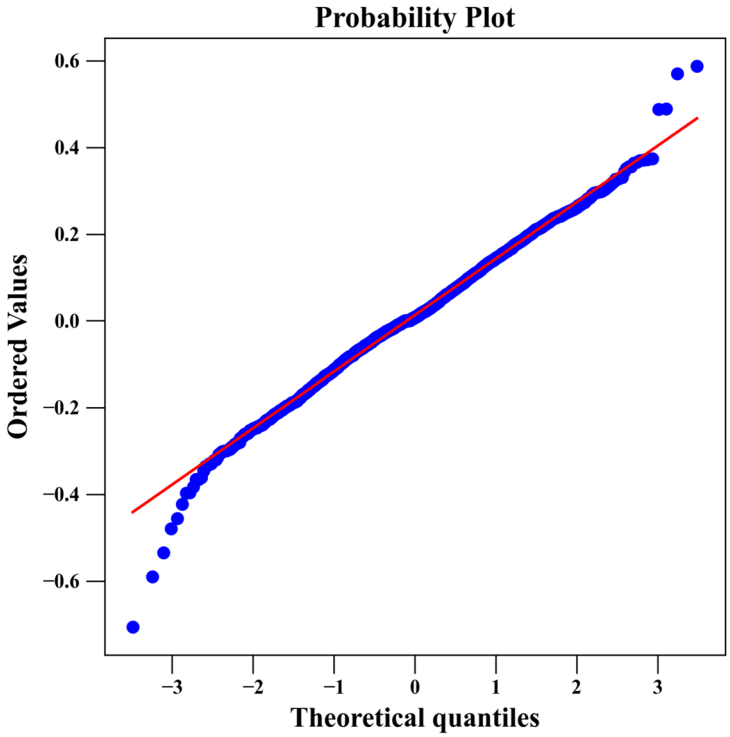


Fig. 7. Q-Q plot evaluating the normality of residuals for the proposed GCN-based readability assessment model on the training set of the CLEAR dataset.

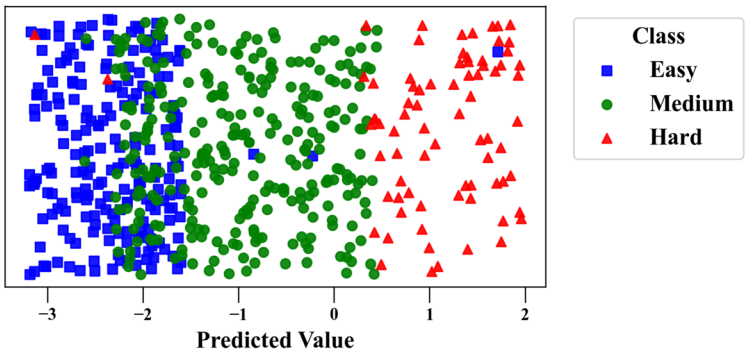


Fig. 8. Swarm plot illustrating the validation of the proposed GCN-based readability assessment model on the Varnamkhasti dataset, showing clusters of predicted readability scores for Persian texts.

MPAA Rating	RMSE	MAE	R^2-score
G	0.160	0.135	0.975
PG	0.170	0.145	0.968
PG-13	0.180	0.150	0.960

Table 3. Performance Across MPAA Rating Subsets on the CLEAR Dataset.

capability. Additionally, the model preserves statistical robustness while capturing complex syntactic and semantic structures. Therefore, it can be used to evaluate the readability of educational content.

Figure 8 swarm plot demonstrates that the proposed graph-based method effectively predicts Persian readability scores on the Varnamkhasti dataset. The plot shows distinct clusters where texts within the same readability class receive similar predicted scores. Using a different language dataset while pre-trained on the CLEAR dataset, these findings validate the proposed method. There are some minor overlaps that suggest some borderline cases, but overall, the method appears robust and adaptable to a wide variety of languages.

Model Variant	RMSE	MAE	R^2
BERT regression (no graph)	0.1917	0.1609	0.9588
Graph features without GCN	0.1781	0.1486	0.9622
GCN without POS weighting	0.1715	0.1440	0.9680
Full model (GCN + POS weighting)	**0.1668**	**0.1406**	**0.9729**

Table 4. Ablation study on the CLEAR dataset to assess the contribution of model components.

Method	RMSE	MAE	R^2-score
Schwarm and Ostendorf[7] (Feature-based)	0.23	0.20	0.91
BERT (Li et al.[16])	0.19	0.16	0.95
ALBERT (Ribeiro et al.[22])	0.18	0.15	0.96
RoBERTa (fine-tuned)	0.17	0.145	0.965
Proposed GCN (Ours)	**0.1668**	**0.1406**	**0.9729**

Table 5. Comparison with Representative Baselines on the CLEAR Dataset.

Analysis of the model's performance across subsets of the CLEAR dataset, categorized by MPAA ratings (G, PG, PG-13), reveals its effectiveness in educational contexts. Table 3 summarizes the results, indicating an R^2 score of 0.975 for G-rated texts, typically simpler and designed for younger learners, compared to 0.968 for PG-rated texts with more complex structures. These findings demonstrate the graph-based approach's capability to capture syntactic nuances in educational texts tailored to varying learner levels, supporting its applicability for classroom text selection.

To isolate the contribution of the proposed GCN-based graph modeling, we conducted an ablation study on the CLEAR dataset (Table 4). A BERT-only regression baseline confirms the strong contribution of pretrained contextual representations. Incorporating graph features without message passing improves performance, indicating that explicit syntactic structure is beneficial. Replacing POS-based edge weighting with uniform weights results in a performance drop, highlighting the importance of differentiated syntactic distance modeling. The full model, combining GCN message passing with POS-aware edge weighting, achieves the best results across all metrics, demonstrating that the gains are attributable to the proposed graph-based architecture rather than to the underlying language model alone.

Further evaluation of the proposed graph-based readability assessment model is conducted by comparing its performance against representative baselines covering traditional feature-based, pretrained language model, and recent neural readability approaches. Specifically, we re-implemented a traditional feature-based model[7], a BERT-based readability predictor inspired by Li et al.[16], and an ALBERT-based variant inspired by Ribeiro et al.[22], and evaluated all methods under the same experimental protocol on the CLEAR dataset. As summarized in Table 5, the proposed GCN-based model achieves an R^2 score of 0.9729, RMSE of 0.1668, and MAE of 0.1406, outperforming the BERT-based model ($R^2 = 0.95$, RMSE = 0.19, MAE = 0.16) and the ALBERT-based model ($R^2 = 0.96$, RMSE = 0.18, MAE = 0.15). The traditional feature-based baseline yields lower performance ($R^2 = 0.91$), indicating the advantage of neural approaches. Compared to pretrained language models that primarily rely on sequential representations, the proposed graph-based framework more effectively captures syntactic structure, which is particularly relevant for modeling readability in educational texts.

While the proposed model demonstrates strong predictive accuracy overall, a closer examination of the residuals and subset performance reveals certain limitations. Notably, the model shows slightly diminished performance when applied to texts with ambiguous or abstract content–particularly those found in literature or social science domains–compared to more structurally rigid STEM passages. This discrepancy likely stems from the syntactic and semantic irregularities common in narrative or philosophical writing, where conventional dependency structures are less consistent. Additionally, shorter texts or sentences with limited syntactic depth present challenges for the graph construction process, as there may be insufficient relational information to capture meaningful complexity. In such cases, the resulting graphs may not fully represent the nuances of the text, leading to less accurate readability predictions. These limitations highlight opportunities for future work to integrate higher-level discourse features, coreference resolution, or hybrid approaches combining graph and transformer-based models to better handle linguistic ambiguity and brevity.

Discussion

The experimental results demonstrate that explicitly modeling syntactic structure through graph representations leads to consistent improvements in readability prediction. Unlike sequential neural models that primarily capture local contextual information, the proposed GCN-based framework enables information propagation across syntactically related tokens, allowing the model to better reflect long-range dependencies that are known to affect sentence processing difficulty.

Although MPAA-style content ratings originate outside formal education, they serve here as coarse-grained proxies for linguistic and thematic complexity rather than as direct indicators of grade level. In the CLEAR

dataset, passages labeled as *General* typically exhibit simple sentence structures, concrete vocabulary, and narrative or introductory expository styles that align with early elementary reading materials. Categories such as *Parental Guidance* and *PG-13* increasingly reflect texts with denser syntax, abstract concepts, and informational or analytical content commonly encountered in middle and secondary school curricula. Higher-rated passages often correspond to advanced academic or argumentative writing characteristic of postsecondary contexts. Viewed through this lens, the distribution shown in Fig. 4 provides insight into how the proposed readability model spans a range of educational text types and subject domains, supporting its relevance for classroom-oriented applications.

The ablation study further reveals that the performance gains cannot be attributed solely to pretrained language representations. While BERT provides a strong semantic foundation, incorporating graph-based message passing yields measurable improvements, indicating that structural information contributes complementary signals beyond lexical semantics. Moreover, the inclusion of POS-based edge weighting enhances performance, suggesting that different intervening syntactic categories impose varying cognitive processing costs, which are relevant to readability.

From an educational perspective, these findings highlight the importance of syntactic complexity in assessing text difficulty, particularly for instructional and learner-facing materials. The proposed approach is therefore well-suited for applications such as curriculum design, automated text adaptation, and readability-aware content generation.

Despite these advantages, the current study has limitations. The model is evaluated on a single dataset and focuses on English texts, which may restrict generalizability to other domains or languages with different syntactic characteristics. Additionally, the computational cost of graph construction and GCN inference may pose challenges for large-scale deployment. Addressing these limitations, as well as extending the framework to multilingual and domain-adaptive settings, constitutes an important direction for future work.

Conclusion

This paper proposed a graph-based readability assessment framework that integrates part-of-speech information with dependency-based syntactic structure, achieving an R^2 score of 0.9729 and an RMSE of 0.1668 on the CLEAR dataset. The model employs Graph Convolutional Networks together with a novel graph construction strategy based on POS-aware syntactic distance, enabling explicit modeling of structural linguistic complexity. Empirical comparisons demonstrate that this formulation consistently outperforms traditional feature-based approaches and transformer-only baselines. By representing words as nodes and syntactic dependencies as weighted edges, the proposed framework moves beyond surface-level indicators and supports a deeper, structure-aware analysis of textual readability.

The proposed model demonstrated strong predictive performance on the CLEAR dataset, achieving an R^2 score of 0.9729 on the test set. Nevertheless, domain-level analysis reveals comparatively lower performance on social science and literature texts than on STEM-oriented materials. This discrepancy suggests that readability challenges in non-STEM domains extend beyond sentence-level syntactic complexity, reflecting a broader range of linguistic and conceptual factors.

A key limitation of the proposed approach lies in its reliance on syntactic graph representations, which may not fully capture the discourse-level coherence, implicit reasoning, and abstract argumentation characteristic of social science and literary texts. Such texts often require pragmatic interpretation and background knowledge that fall outside the scope of purely structural modeling. This limitation underscores an important consideration for educational deployment: automated readability scores should be interpreted as decision-support signals rather than definitive assessments, particularly in classroom contexts where pedagogical intent and learner background are critical. Addressing these challenges by incorporating discourse-aware representations or domain-adaptive mechanisms constitutes a promising direction for future work.

The model was also evaluated on a classification-based dataset to assess its ability to generate consistent readability scores for texts within the same category. The results show that the method generalizes well across different readability levels and languages, outperforming traditional readability metrics. Furthermore, the integration of Bayesian Optimization allowed for efficient hyperparameter tuning, enhancing the model's robustness. Overall, the proposed method provides accurate readability predictions and offers a scalable, adaptable solution suitable for diverse educational contexts.

Future work can explore expanding the model to incorporate multilingual datasets, further refining syntactic and semantic feature extraction, and integrating user feedback to enhance adaptability. Other graph representation learning methods can be used in addition to the proposed graph construction method. Also, while BERT is used in this work, the proposed graph-based readability assessment framework is model-agnostic and can be extended to incorporate alternative pretrained language models, such as RoBERTa or ALBERT, which we leave for future investigation.

Data availability

The datasets used during the current study are available from the corresponding author on reasonable request.

Code availability

The custom code used for syntactic graph construction and GCN-based readability modeling is not publicly available. To ensure reproducibility, all algorithmic details, hyperparameter settings, and experimental procedures are fully described in the Methods section. Access to the code for academic, non-commercial purposes may be provided by the corresponding author upon reasonable request.

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Author contributions

L.Z. conceived the study, designed the methodology, implemented the proposed model, conducted the experiments, performed the data analysis, and wrote the original draft of the manuscript. J.A., J.B., A.Y., M.S.A.Y., F.M., B.B., P.K.S., and S.V. contributed to the literature review, experimental design discussions, and interpretation of the results. J.A., J.B., and A.Y. assisted in refining the methodological framework and validating the experimental setup. F.M., B.B., and P.K.S. contributed to manuscript review, editing, and technical revisions. S.V. supported result analysis and visualization. All authors reviewed, revised, and approved the final version of the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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